7.5.4 Rotation of Axes; Second Degree Equations

So far, all major/minor axes, transverse/conjugate axes, and directrixes have been vertical or horizontal. We now consider situations where this is not the case.

Example 1:

Consider $xy = 1 \Leftrightarrow y = \frac{1}{x}$ (it is a hyperbola). Rotating the *x* and *y* axes by 45° makes it easier to analyze this hyperbola.

General Idea:

Theorem: Let θ be the angle of rotation of the x and y axes.

Note: We can see that this theorem does indeed work if we consider example 1.

Example 2:

Find the new coordinates of the point $(2,4)$ if the new coordinates axes are rotated through an angle of $\theta = 30^\circ$.

Example 3:

Express $xy = 1$ in terms of new $x'y'$-coordinates by rotating the axes through an angle of $\theta = 45^\circ$

Theorem: If a conic is given by the equation

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

Proof: see textbook pages 750&751.

Note: We will always choose θ to be acute.

Example 4: Identify and sketch the conic given by the equation

$$x^2 + \sqrt{3}xy + 2y^2 - 10 = 0$$

Example 5: Consider the conic given by the equation

$$153x^2 - 192xy + 97y^2 - 30x - 40y - 200 = 0$$

Example 6: Show that the given equation is a parabola, then find its vertex, focus, and directrix in terms of the *xy*-axes and sketch a graph.

$$\frac{1}{2}\left(x^2+y^2\right)-2\left(x+y-1\right)-xy=0$$